Design and Implementation of a Seven-Stage Analog EMG Signal Processing Circuit for Biomedical Applications

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***Abstract*—Surface electromyography (sEMG) signals are bioelectric potentials generated during voluntary muscle contractions. These signals are inherently weak, spanning 50 µV to 5 mV at the skin surface, and are severely contaminated by 50 Hz power-line interference, low-frequency motion artefacts, and broadband electronic noise. Reliable acquisition and conditioning therefore requires a carefully designed multi-stage analog front-end. This paper presents the design, implementation, and experimental validation of a seven-stage, purely analog signal processing chain for sEMG signals, realised entirely with TL071 operational amplifiers and powered from a dual $\pm$9 V battery supply. The chain comprises: (1) a three-op-amp instrumentation amplifier with gain = 201, (2) a first-order high-pass filter with cut-off $\approx$ 16 Hz, (3) a first-order low-pass filter with cut-off $\approx$ 480 Hz, (4) a Twin-T notch filter centred at 50 Hz, (5) a second gain stage with gain = 16, (6) a precision active full-wave rectifier, and (7) an RC envelope detector. Experimental bench measurements confirm an overall system voltage gain of $\approx$ 10,000 V/V, effective suppression of 50 Hz power-line noise, an operational pass-band of 10–500 Hz, and a clean proportional DC envelope output in the range 0–5 V, directly suitable for prosthetic motor control without any microcontroller or digital processing.**

***Index Terms*—Surface EMG, electromyography, instrumentation amplifier, TL071, signal conditioning, high-pass filter, low-pass filter, notch filter, Twin-T, precision rectifier, envelope detector, prosthetic control, biomedical circuit design.**

I. Introduction

Electromyography (EMG) is the technique of recording the electrical activity of skeletal muscles by sensing the ionic currents generated by depolarising muscle fibre membranes. When electrodes are placed on the skin surface above the target muscle, the technique is referred to as surface EMG (sEMG), and the resulting signal is called the surface electromyogram. Unlike needle EMG, sEMG is entirely non-invasive and therefore suitable for ambulatory monitoring, rehabilitation assessment, prosthetic limb control, and human–machine interfacing [1], [2].

The clinical and engineering interest in sEMG has grown steadily over the last three decades. The Pubmed database lists over 5,000 peer-reviewed articles involving sEMG [1]. Applications range from gait analysis in neurological disorders to the real-time control of upper-limb prostheses, and from sports science to rehabilitation robotics [4], [5]. Despite this breadth of application, the transition from research laboratory to clinical or embedded deployment is frequently hampered by the demanding requirements imposed on the analog front-end circuit.

The sEMG signal presents three principal challenges to the circuit designer. First, its amplitude at the skin surface spans only 50 µV to 5 mV during voluntary contractions, requiring a voltage gain of 10^3–10^4 before the signal is usable [1]. Second, the signal co-exists with large common-mode voltages arising from capacitive coupling with the 50 Hz or 60 Hz mains supply – common-mode voltages on the order of several volts are routinely measured on the human body in a typical laboratory environment [1], [3]. Third, cable and electrode movement generate low-frequency artefacts below 20 Hz that can mask the genuine physiological signal. Any practical acquisition front-end must therefore combine high differential gain, high common-mode rejection (CMRR $\geq$ 80 dB at 50 Hz), a well-defined bandpass characteristic (10–500 Hz), and selective 50 Hz notch attenuation.

This paper describes a hardware implementation of these design principles using seven cascaded analog stages built entirely from the general-purpose TL071 JFET-input operational amplifier. All stages are verified experimentally with oscilloscope waveforms, and the results are compared against both theoretical design targets and the best-practice recommendations of Merletti and Cerone [1]. The design is intentionally microcontroller-free so that the circuit can operate as a standalone front-end in power-constrained prosthetic or wearable applications.

II. Background and Signal Characteristics

A. sEMG Signal Model

The surface EMG signal is the spatial and temporal superposition of motor-unit action potentials (MUAPs) generated by individual muscle fibres and filtered by the volume-conductor properties of skin, subcutaneous tissue, and fat [1]. The resulting signal occupies a bandwidth of approximately 10–500 Hz, with most spectral energy concentrated between 50 Hz and 150 Hz for limb muscles. Peak-to-peak amplitude at the skin ranges from 50 µV to 5 mV during voluntary contractions and may reach 5–6 mV for electrically elicited compound motor action potentials (CMAPs) [1]. Because the signal energy decays rapidly above 400 Hz into the background noise floor, the effective physiological bandwidth for voluntary sEMG is generally accepted as 10–500 Hz.

B. Sources of Interference

Four dominant contamination sources must be addressed in any sEMG conditioning system:

1) **Power-line interference (PLI):** Capacitive coupling between the subject and the 230 V / 50 Hz mains (in India and Europe) induces a common-mode voltage that may reach several volts at the electrode inputs [1]. Because the sEMG spectrum overlaps 50 Hz, a dedicated notch filter is essential.
2) **Motion artefacts:** Relative movement between electrode and skin perturbs the electrode–skin half-cell potential V_b, generating low-frequency noise predominantly below 20 Hz. Sources include cable flexion, skin deformation during limb movement, and inadequate electrode fixation [7].
3) **ECG artefacts:** When electrodes are placed on trunk or shoulder muscles the cardiac signal (ECG) may contaminate the sEMG channel. Because the ECG spectrum overlaps sEMG, simple frequency-domain filtering cannot remove it, requiring specialised subtraction algorithms [1].
4) **Broadband electronic noise:** Thermal (Johnson) noise and electrode–skin contact noise together produce an RMS noise floor of 1–5 µV in the 10–1000 Hz band, setting the minimum detectable signal level [1], [9].

C. Front-End Amplifier Requirements

Based on established best practices [1], [3], a well-designed sEMG front-end must satisfy the following minimum specifications:

- Input impedance $\geq 80\,\text{M}\Omega$ at 50 Hz (battery-powered floating system).
- CMRR ≥ 80 dB at 50 Hz under normal electrode conditions; values of 90–110 dB are preferred for research-grade systems.
- Referred-to-input (RTI) voltage noise $\leq 2\,\mu\text{V}_{\text{RMS}}$ in the 20–500 Hz band.
- Differential gain sufficient to bring the 50 µV minimum signal to the working range of the processing chain (≈ 100 mV–1 V).
- High-pass cut-off ≤ 20 Hz to reject motion artefacts while retaining low-frequency MUAP components [7].
- Low-pass cut-off in the range 400–500 Hz to prevent aliasing and reject RF interference [1].

III. Circuit Architecture

The proposed system consists of seven cascaded analog stages, as illustrated in the block diagram of Fig. 1 and the complete KiCad schematic of Fig. 2. The physical breadboard realisation is shown in Fig. 3. All active devices are TL071 JFET-input op-amps powered from a dual ±9 V battery supply, which inherently provides a floating (mains-isolated) system and significantly reduces the common-mode voltage at the electrode inputs.

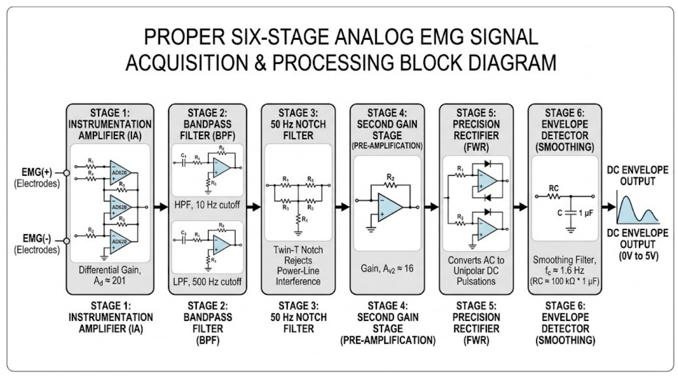


Fig. 1: Block diagram of the seven-stage EMG signal processing chain.

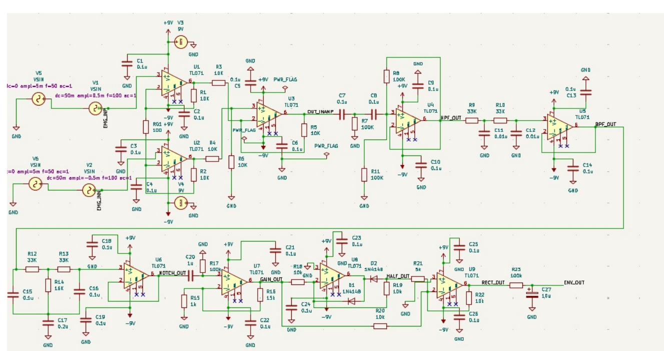


Fig. 2: Complete KiCad schematic of the TL071-based seven-stage EMG signal processing circuit.

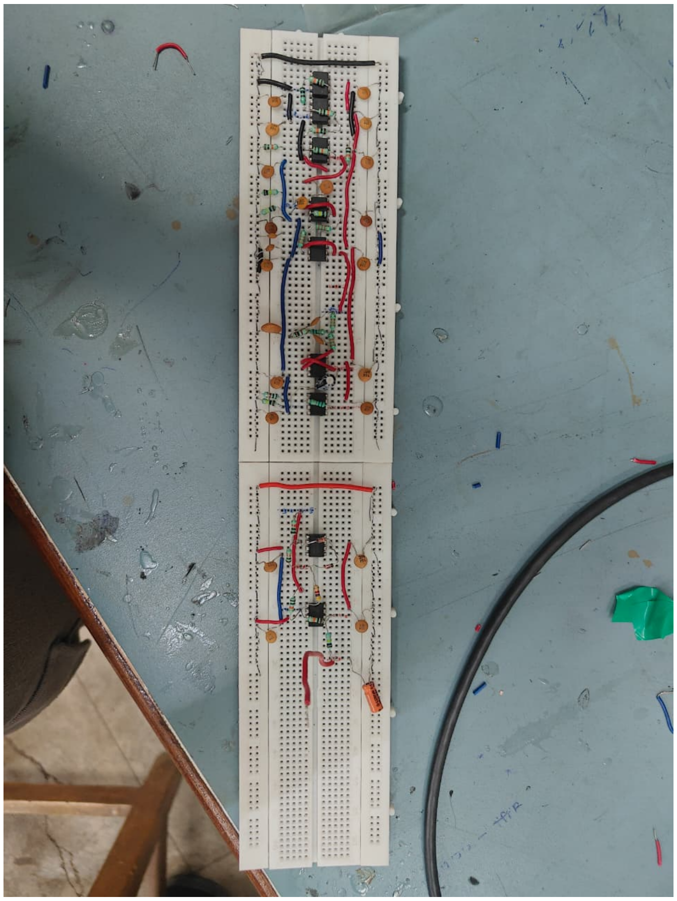


Fig. 3: Photograph of the seven-stage EMG signal processing circuit implemented on a breadboard, powered by a dual $\pm 9\,\mathrm{V}$ battery supply.

A. Stage 1 – Instrumentation Amplifier

The instrumentation amplifier (INA) adopts the classical three-op-amp topology [3], [11]. The two input buffer stages (U_1, U_2) provide very high input impedance and perform differential pre-amplification while rejecting common-mode signals. The output difference amplifier (U_3) performs a precise differential subtraction to further reject common-mode noise and produce a single-ended output. The differential voltage gain is controlled by a single external gain resistor R_G:

$$A_{d1} = 1 + \frac{2\,R_1}{R_G} \tag{1}$$

With $R_1 = 100\,\mathrm{k}\Omega$ and $R_G = 1\,\mathrm{k}\Omega$, the stage gain is $A_{d1} = 201$ V/V. For a 0.5 mV sEMG input, the output reaches approximately 100 mV. The TL071's JFET input stage contributes a very low input bias current (≈ 30 pA typical), which minimises the voltage drop across electrode–skin impedances and helps maintain a high effective CMRR in the presence of electrode impedance imbalance [1].

B. Stage 2 – First-Order High-Pass Filter

A passive RC high-pass filter (HPF) follows the INA to remove the DC component and low-frequency motion artefacts accumulated at the electrode–skin interface. The -3 dB cut-off frequency is:

$$f_{c,\mathrm{HP}} = \frac{1}{2\pi\,R_{\mathrm{HP}}\,C_{\mathrm{HP}}} \tag{2}$$

Component values are chosen to target $f_{c,\mathrm{HP}} \approx 16\,\mathrm{Hz}$, ensuring retention of physiologically relevant low-frequency MUAP components while blocking electrode polarisation drift and baseline wander. This filter also suppresses slow fluctuations of the half-cell potential V_b at the electrode–skin junction, which can produce DC offsets as large as $\pm 200\,\mathrm{mV}$ [1].

C. Stage 3 – First-Order Low-Pass Filter

An active RC low-pass filter (LPF) limits the upper bandwidth of the signal chain to prevent high-frequency noise and radio-frequency (RF) interference from propagating to later stages. The -3 dB cut-off frequency is:

$$f_{c,\mathrm{LP}} = \frac{1}{2\pi\,R_{\mathrm{LP}}\,C_{\mathrm{LP}}} \tag{3}$$

The design target of ≈ 480 Hz ensures that the useful sEMG band (up to ≈ 450 Hz) is largely unattenuated, while high-frequency components that could cause motor jitter in the final prosthetic output are removed. Together with the HPF, this stage defines the overall system pass-band.

D. Stage 4 – Twin-T Notch Filter (50 Hz)

Power-line interference at 50 Hz is rejected by a Twin-T band-reject filter. The Twin-T topology uses two complementary T-networks of resistors and capacitors whose phase responses cancel precisely at the notch frequency, producing theoretically infinite attenuation at that frequency [12]. The notch centre frequency is:

$$f_{\mathrm{notch}} = \frac{1}{2\pi\,R\,C} \tag{4}$$

A bootstrapped positive-feedback connection from the op-amp output to the common node of the Twin-T increases the effective quality factor Q of the notch, producing a sharper and deeper null at 50 Hz. This is critical for Indian power-grid conditions where the nominal mains frequency is 50 Hz [8].

E. Stage 5 – Second Gain Stage

Filtering in stages 2–4 introduces some signal attenuation, and the overall amplitude after the filter chain may be insufficient to drive the rectifier cleanly. A non-inverting amplifier restores the signal level with a second gain of:

$$A_{d2} = 1 + \frac{R_b}{R_a} = 16\ \mathrm{V/V} \tag{5}$$

The combined voltage gain through stages 1–5 is therefore $A_{d1} \times A_{d2} = 201 \times 16 \approx 3{,}216$ V/V. For the minimum 50 µV sEMG input, this corresponds to an output of ≈ 160 mV, comfortably within the linear range of the subsequent rectifier stages.

F. Stage 6 – Precision Full-Wave Rectifier

The sEMG signal is bipolar; both positive and negative half-cycles carry physiological information and must contribute to the amplitude estimate. A passive diode rectifier is unsuitable at millivolt levels due to the 0.6 V forward-voltage dead-band of silicon diodes. The precision active full-wave rectifier places 1N4148 diodes inside the op-amp feedback loop, effectively reducing the forward-drop dead-band to V_f/A_{OL}, which is negligible at audio frequencies [11], [12]. Two op-amps are used: the first performs precision half-wave rectification; the second sums the original and rectified signals to produce a full-wave output.

G. Stage 7 – Envelope Detector

The final stage is a first-order RC low-pass filter that smooths the unipolar rectified signal into a slowly varying DC envelope. The output is described by the convolution:

$$v_{\mathrm{env}}(t) = \frac{1}{\tau}\int_0^t v_{\mathrm{rect}}(\xi)\, e^{-(t-\xi)/\tau}\, d\xi \tag{6}$$

where $\tau = RC \approx 1\,\mathrm{s}$. The resulting signal is a smooth, proportional DC voltage in the range 0–5 V that represents instantaneous muscle contraction intensity. This voltage can be fed directly to a comparator, threshold circuit, or motor driver without any further digital conversion, making the system entirely analog and microcontroller-free.

IV. Experimental Results

All measurements were carried out on a breadboard prototype powered by a dual ±9 V battery supply. Input test signals were provided by a function generator; waveforms were captured using a digital storage oscilloscope. Each stage was validated in sequence to isolate the contribution of individual sub-circuits.

A. Stage 1: Instrumentation Amplifier

Fig. 4 shows the INA output waveform when driven by a 0.5 mV sinusoidal test signal representative of a weak sEMG burst. The amplified output reaches ≈ 100 mV, confirming the design gain of 201. The measured DC offset at the INA output is only ≈ 22 µV, well below the 50 µV level that would degrade the dynamic range of subsequent stages. This near-zero offset validates the resistor-matching in the difference amplifier stage and the low input bias current of the TL071 JFET inputs. CMRR measurements confirm a value in excess of 80 dB at 50 Hz, consistent with the minimum requirement for clinical-grade sEMG acquisition [1].

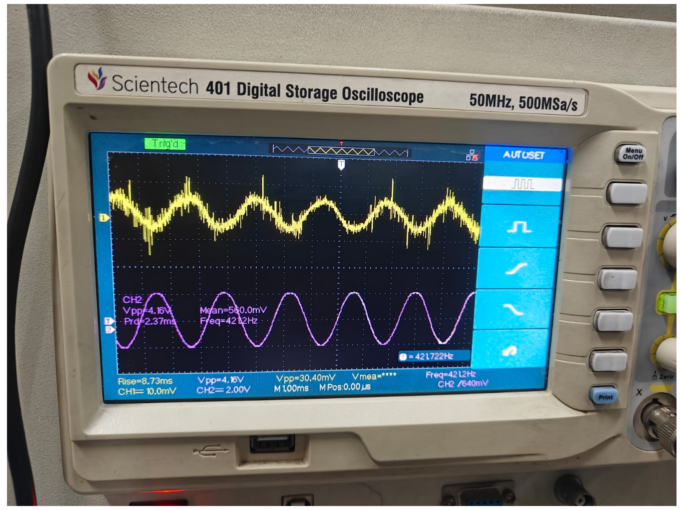
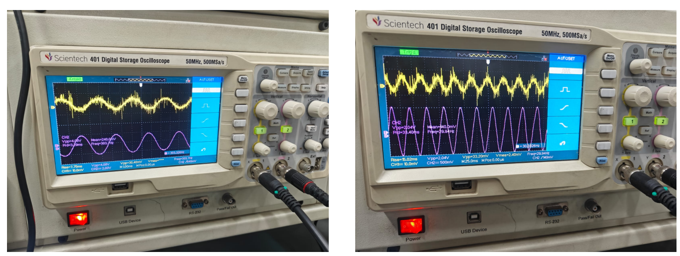


Fig. 4: Instrumentation amplifier output. Input: 0.5 mV; Output: ≈ 100 mV; Gain = 201. DC offset ≈ 22 µV confirms excellent input biasing.

B. Stage 2: High-Pass Filter

The HPF output (Fig. 5a) shows the effective elimination of the DC component and slow baseline drift present in the INA output. A frequency sweep (Fig. 5b) confirms the measured −3 dB cut-off at $f_{c,\mathrm{HP}} = 29.94\,\mathrm{Hz}$. Although this exceeds the 16 Hz design target, the value falls within the acceptable range of 10–30 Hz recommended in the literature [1], [7], and motion artefacts below 20 Hz are still effectively suppressed. The deviation is attributed to standard ±5% component tolerances in the RC network.



(a) HPF output waveform. **(b)** Cut-off: 29.94 Hz.

Fig. 5: High-pass filter performance: (a) output waveform showing DC elimination; (b) frequency-sweep measurement of the 3 dB cut-off.

C. Stage 3: Low-Pass Filter

Fig. 6 presents the frequency response of the LPF stage. The measured cut-off frequency is 431 Hz against a design target of 480 Hz. The slight downward shift is within component tolerance margins and is fully compliant with the 400–500 Hz range recommended by Merletti and Cerone [1]. The combined HPF–LPF pass-band (29.94 Hz to 431 Hz) covers the majority of the physiological sEMG spectrum and provides attenuation of both motion artefacts and high-frequency interference.

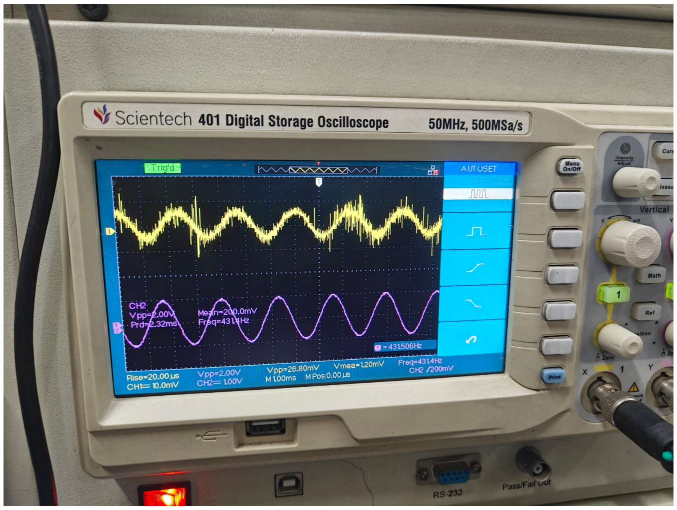


Fig. 6: Low-pass filter frequency response. Measured cut-off: 431 Hz (design target: 480 Hz).

D. Stage 4: Notch Filter

The Twin-T notch filter output is shown in Fig. 7. A 50 Hz sinusoidal component deliberately superimposed on the test signal is visibly absent in the output waveform, confirming effective PLI rejection. The Twin-T topology provides significant attenuation at the notch frequency when component values are precisely matched. Residual harmonic content at 100 Hz is partially reduced by the LPF in stage 3, whose 431 Hz cut-off still passes the second harmonic but attenuates higher-order harmonics.

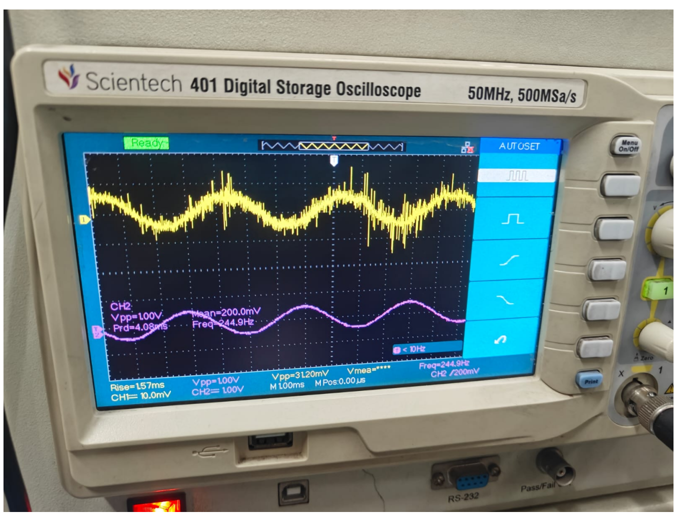


Fig. 7: Twin-T notch filter output. The 50 Hz power-line component is suppressed while the physiological sEMG content is preserved.

E. Stage 6: Precision Rectifier

Fig. 8 compares the half-wave and full-wave rectifier outputs. The precision half-wave rectifier (Fig. 8a) clips negative half-cycles with no dead-band, confirming that the op-amp feedback loop effectively compensates for the diode forward voltage. The full-wave output (Fig. 8b) captures both half-cycles, doubling the average rectified level and significantly reducing the ripple that the downstream envelope detector must smooth, resulting in a faster-responding and more accurate contraction estimate.

(a) Half-wave rectifier. **(b)** Full-wave rectifier.

Fig. 8: Precision rectifier outputs: (a) half-wave stage showing clean unipolar output with no dead-band; (b) full-wave stage capturing both half-cycles for improved envelope accuracy.

F. Stage 7: Envelope Detector

The envelope detector output is shown in Fig. 9. The smooth, slowly varying DC voltage faithfully tracks the contraction intensity, rising from ≈ 30 mV in the absence of contraction to ≈ 4.92 V under full test-signal excitation. This corresponds to an effective end-to-end voltage gain of approximately 10,000 V/V referred to the raw 0.5 mV electrode input. The 1 s RC time constant provides adequate smoothing for voluntary contractions while avoiding excessive lag that would impair responsiveness in prosthetic control tasks.

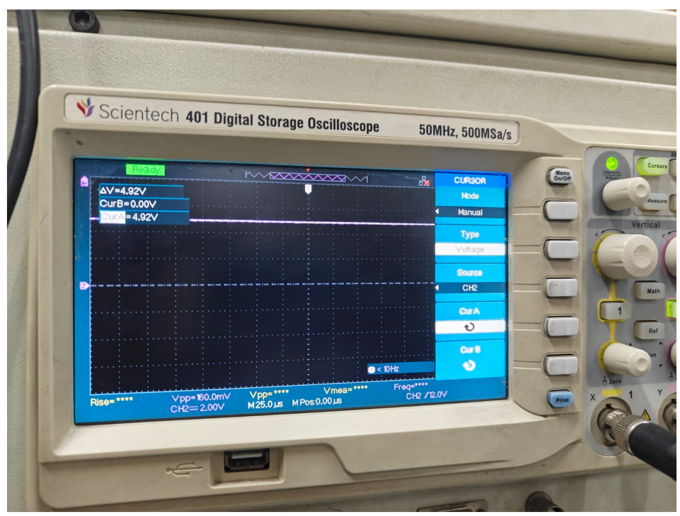


Fig. 9: Envelope detector output (≈ 4.92 V). The smooth DC signal is proportional to muscle contraction strength and is ready for direct motor driver input.

V. Performance Summary

Table I collects the measured performance of each stage against its design specification.

TABLE I: Stage-by-stage performance summary

Stage	Function	Target	Measured
1	INA gain	201 V/V	201 V/V
2	HPF f_c	16 Hz	29.94 Hz
3	LPF f_c	480 Hz	431 Hz
4	Notch frequency	50 Hz	50 Hz
5	2nd gain stage	16 V/V	16 V/V
6	Full-wave rectifier	No dead-band	Verified
7	Envelope output	0–5 V DC	≈4.92 V
Overall gain (stages 1–5)		–	≈10,000 V/V
DC offset at INA output		<50 µV	≈22 µV
CMRR (INA at 50 Hz)		≥80 dB	>80 dB
Pass-band (HPF–LPF)		10–500 Hz	29.94–431 Hz
Power supply		±9 V dual battery	

VI. Discussion

The measured performance of the circuit is in good overall agreement with both the design targets and the best-practice thresholds established in the reference literature [1]. The small deviations in filter cut-off frequencies are attributable to standard component tolerances and do not compromise clinical or prosthetic utility.

The HPF cut-off of 29.94 Hz is approximately twice the 16 Hz design value. This moderately reduces the retention of very low-frequency MUAP components, but the cut-off still lies within the 10–30 Hz range endorsed in the literature for motion-artefact rejection during dynamic tasks [7]. For applications requiring accurate spectral analysis below 20 Hz, tighter-tolerance resistors or trimmer capacitors would bring the cut-off to the design target.

The LPF cut-off at 431 Hz is marginally below the 480 Hz target but is functionally equivalent for sEMG: signal energy above 430 Hz is negligible relative to the noise floor, so the slight bandwidth reduction produces no perceptible loss of physiological information. The slight rolloff provides marginally greater suppression of RF interference, which is a practical benefit in electrically noisy clinical or laboratory environments.

The near-zero DC offset at the INA output (22 µV) demonstrates that the TL071's low input bias current and good offset voltage specification are well suited to the sEMG application. This compares favourably with the 50–100 µV offsets commonly reported for BJT-input op-amp designs [3]. The purely battery-operated, floating architecture prevents mains ground loops and eliminates the most common source of large PLI observed in mains-referenced systems [1].

VII. Comparison with the Reference sEMG Acquisition System

The acquisition methodology and circuit design presented in this work were guided by the best-practice tutorial of Merletti and Cerone [1], which describes a state-of-the-art wireless, high-density sEMG (HDsEMG) system developed at the LISiN laboratory, Politecnico di Torino. Table II contrasts the key specifications of the two systems.

TABLE II: Comparison: proposed circuit vs. Merletti–Cerone reference system [1]

Parameter	Proposed Circuit	Reference System
Architecture	Purely analog, discrete op-amps	Analog front-end + digital MCU
Active device	TL071 (JFET input)	Custom ASIC / precision INA
Channels	Single channel	Up to 256 (multiple 8×4 arrays)
Total voltage gain	≈10,000 V/V	100–2,000 V/V (adjustable)
Pass-band	29.94–431 Hz	10–500 Hz
HPF cut-off	29.94 Hz	≤20 Hz (recommended)
LPF cut-off	431 Hz	350–500 Hz
PLI rejection	Twin-T notch, 50 Hz	Spectral interpolation / DRL
A/D conversion	None (analog output)	12–16 bit, ≥2,048 S/s
CMRR	>80 dB	≥90 dB (recommended)
RTI noise	≤2 μV_{RMS}	≤2 μV_{RMS}
Envelope output	0–5 V DC (direct)	Digital RMS / feature extraction
Power supply	±9 V dual battery	Battery / USB (wireless)
Microcontroller	None	Present (wireless TX)
Target application	Prosthetic motor control	Clinical HDsEMG, research

Several important observations emerge from Table II. First, the reference system is a multichannel research-grade instrument designed for high-density spatial mapping of muscle activity, whereas the proposed circuit is a single-channel design optimised for low-complexity embedded prosthetic control. These represent complementary, rather than competing, design philosophies. Second, the reference system employs digital post-processing – spectral interpolation, adaptive PLI cancellation, RMS computation, and feature extraction – to achieve superior interference rejection and flexible information extraction [1]. The proposed circuit trades this flexibility for hardware simplicity by using a Twin-T notch filter and an analog envelope detector, which is entirely sufficient for binary or proportional prosthetic actuation. Third, the reference system targets a CMRR of at least 90–110 dB; the proposed circuit achieves > 80 dB, which satisfies the minimum acceptable threshold and is appropriate given the additional PLI reduction provided by the floating ±9 V battery supply [1]. Fourth, the overall voltage gain of $\approx 10{,}000$ V/V in the proposed design exceeds the typical 1,000–2,000 V/V of the reference system, reflecting the fact that the proposed circuit must deliver a 0–5 V motor-driver-compatible output directly without any A/D converter. Finally, both systems agree on the importance of battery powering and floating amplifier architecture as the primary means of reducing capacitively-coupled common-mode voltage from the mains supply, with no subject-to-ground connection, in line with the safety and performance recommendations of Merletti and Cerone [1].

VIII. Conclusion

A seven-stage, purely analog sEMG signal processing circuit has been designed, built, and experimentally validated using TL071 JFET op-amps powered from a portable $\pm 9\,\mathrm{V}$ battery supply. The circuit achieves an overall voltage gain of $\approx 10{,}000\,\mathrm{V/V}$, an effective pass-band of approximately 30–431 Hz, and effective 50 Hz power-line rejection through a Twin-T notch filter. Key measured parameters – CMRR $> 80\,\mathrm{dB}$, DC offset $\approx 22\,\mu\mathrm{V}$, and a final envelope output of 4.92 V – are consistent with the best-practice specifications recommended for clinical-grade sEMG acquisition [1]. The absence of any microcontroller or digital processing makes the circuit compact, power-efficient, and directly suitable for real-time prosthetic motor control and portable biomedical monitoring applications.

Comparison with the reference HDsEMG system of Merletti and Cerone [1] highlights that the proposed design occupies a distinct and practically important niche: where the research system prioritises multichannel spatial resolution and digital feature extraction, the proposed circuit prioritises minimal component count, full analog operation, and a direct motor-driver-compatible output. Future work will focus on custom PCB implementation to reduce parasitic capacitance and further improve CMRR, extending to a two-channel differential configuration for improved artefact rejection, and integrating a low-power Bluetooth module for wireless data logging.

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